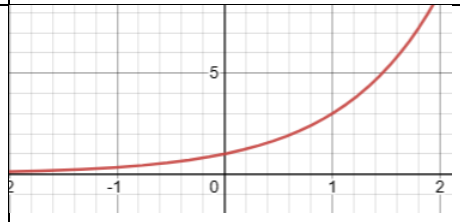
Question		Answer	Marks	AO	Guidance
15	(a)		B1 B1 [2]	1.1 1.1	correct shape in both quadrants; may touch, but not cut negative x-axis do not condone another asymptote drawn. unique intercept at (0,1) identified on graph or independently
15	(b)	$3^{2x} - 6 \times 3^x + 12 = 3^x$ $(3^x)^2 - 7(3^x) + 12 = 0$ $(3^x - 3)(3^x - 4) = 0$ $x = 1$ and $x = \log_3 4$ or $\frac{\log 4}{\log 3}$ or $\frac{\ln 4}{\ln 3}$ found (1, 3) and ($\log_3 4$ **oe**, 4)	M1* M1 M1 A1 A1 [5]	3.1a 1.1 1.1 1.1 3.2a	Three term quadratic structure recognised; may be implied by later work. May see substitution **FT** solving their quadratic. May use formula or calculator. May be in terms of u or another variable. If **M1M1M1A0A0** allow **SCA1** for (1, 3) *or* ($\log_3 4$ **oe**, 4)